VEDA ACADEMIC ASSESSMENT SHEET

SUBJECT: CS | **MAX MARKS: 20 PTS**
CLASS/GRADE: DEV | **DATE: 29/5/2026**

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STUDENT NAME: ______________________ **ROLL NO / ID:** __________________

SECTION A: OBJECTIVE ASSESSMENT

Instruction: Choose the correct option below. Each question carries equal designated points.

Q1. Which of the following scenarios will cause a Java 21 Virtual Thread to pin its carrier thread (underlying platform thread), potentially causing a performance bottleneck? **(10 Pts)**

Q2. Consider the following Java code: **(10 Pts)**

```
class Parent {
   static {
      System.out.print("A");
   }
}
class Child extends Parent {
   static {
      System.out.print("B");
   }
   public static final int MAX = 100;
}
```

What is printed when executing System.out.println(Child.MAX); assuming the classes were not previously loaded?

Q3. Given the class declaration 'class MyNode extends Node<Integer>', where Node<T> defines a method 'void setData(T data)'. What is the purpose of the synthetic 'bridge' method generated by the Java compiler in the compiled bytecode of MyNode? **(10 Pts)**

Q4. Under the Java Memory Model (JMM), which of the following guarantees is NOT provided by declaring a field as 'volatile'? **(10 Pts)**

Q5. How does the Z Garbage Collector (ZGC) achieve ultra-low pause times (typically sub-millisecond) even with multi-terabyte heaps? **(10 Pts)**

Q6. When using 'CompletableFuture.supplyAsync(task, executor)', what happens if an unhandled runtime exception is thrown inside the task? **(10 Pts)**

Q7. What is the primary architectural hazard of invoking 'Stream.parallel()' on a stream that depends on a stateful lambda expression? **(10 Pts)**

Q8. In a 64-bit JVM, what is the primary mechanism and limitation of 'Compressed OOPs' (Ordinary Object Pointers)? **(10 Pts)**

Q9. In Java 8+, HashMap converts a linked list bucket to a balanced tree (treeification) when the chain reaches 8 elements. Which secondary condition must also be met for this treeification to happen, preventing premature resizes? **(10 Pts)**

Q10. During custom object deserialization, which of the following method signatures can be implemented to replace the deserialized object with an alternate instance (e.g., to preserve a Singleton pattern)? **(10 Pts)**